ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE B · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Electronics and Digital Concepts for Robotics paper 1 · practice b.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and neat, recognisable sketches.

SECTION A

QUESTION 1

1.1 1.1.1 B ✓ 1.1.2 A ✓ 1.1.3 C ✓ 1.1.4 B ✓ 1.1.5 D ✓ 1.1.6 C ✓ 1.1.7 B ✓ 1.1.8 D ✓ 1.1.9 A ✓ 1.1.10 C ✓ (10)

1.2 1.2.1 False ✓ 1.2.2 True ✓ 1.2.3 True ✓ 1.2.4 True ✓ 1.2.5 True ✓ 1.2.6 False ✓ 1.2.7 False ✓ 1.2.8 True ✓ 1.2.9 False ✓ 1.2.10 False ✓ (10)

1.3 1.3.1 C ✓ 1.3.2 F ✓ 1.3.3 E ✓ 1.3.4 I ✓ 1.3.5 J ✓ 1.3.6 G ✓ 1.3.7 A ✓ 1.3.8 D ✓ 1.3.9 H ✓ 1.3.10 B ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1.1 The force that drives magnetic flux through a magnetic circuit; mmf = N × I, in ampere-turns (At) ✓ (1)

2.1.2 The opposition of a magnetic circuit to the flux in it (like resistance in an electric circuit), in At/Wb ✓ (1)

2.2.1 $F_m = N \times I$ ✓ $= 400 \times 2{,}5 = 1\,000$ At ✓ (2)

2.2.2 $H = F_m \div l$ ✓ $= 1\,000 \div 0{,}25 = 4\,000$ At/m ✓ (2)

2.3 Faraday: an emf is induced in a conductor when the magnetic flux linking it changes, and its size is proportional to the rate of change of flux ✓. Lenz: the induced emf (current) always opposes the change that produces it ✓. (2)

2.4 $e = N \times \Delta\Phi \div \Delta t$ ✓ $= 200 \times (0{,}05 - 0{,}01) \div 0{,}2$ ✓ $= 40$ V ✓ (3)

[11]

QUESTION 3

3.1.1 $N_s = N_p \times V_s \div V_p$ ✓ $= 1\,150 \times 15 \div 230 = 75$ turns ✓ (2)

3.1.2 Turns ratio $= N_p : N_s = 1\,150 : 75 = 15{,}33 : 1$ ✓ (1)

3.1.3 $I_s = V_s \div R$ ✓ $= 15 \div 6 = 2{,}5$ A ✓ (2)

3.1.4 $I_p = I_s \times V_s \div V_p$ ✓ $= 2{,}5 \times 15 \div 230 = 0{,}16$ A ✓ (2)

3.2.1 $E = 6 \times 1{,}5 = 9$ V ✓ (1)

3.2.2 $r = 6 \times 0{,}2 = 1{,}2\ \Omega$ ✓ (1)

3.2.3 $I = E \div (R + r)$ ✓ $= 9 \div (8{,}8 + 1{,}2) = 0{,}9\ A$ ✓ (2)

3.2.4 $V = I \times R = 0{,}9 \times 8{,}8 = 7{,}92\ V$ ✓ (1)

3.3 Any TWO, each with a correct application ✓✓: series – starting motors / cranes / traction (high starting torque); shunt – lathes, fans, conveyors (nearly constant speed); compound – presses, lifts, rolling mills; separately excited – speed-controlled drives. (2)

[14]

QUESTION 4

4.1 $R = \rho l \div A$ ✓ $= (1{,}72 \times 10^{-8} \times 50) \div (1{,}5 \times 10^{-6})$ ✓ $= 0{,}57\ \Omega$ ✓ (3)

4.2 Forward bias: P side to positive, N side to negative ✓; the depletion layer narrows and the diode conducts once about 0,7 V (silicon) is reached ✓. Reverse bias: P side to negative ✓; the depletion layer widens and only a tiny leakage current flows (until breakdown) ✓. (4)

4.3 Base, collector, emitter ✓. A small base current from the microcontroller pin (through a base resistor) ✓ drives the transistor into saturation so that it conducts from collector to emitter ✓ and the relay coil in the collector circuit is energised; a flywheel diode across the coil protects the transistor when it switches off ✓. (4)

4.4.1 $A_v = -R_f \div R_{in}$ ✓ $= -47 \div 4{,}7 = -10$ ✓ (2)

4.4.2 $V_{out} = A_v \times V_{in} = -10 \times 0{,}3 = -3\ V$ ✓ (1)

4.5 $1/C_T = 1/10 + 1/20 + 1/20 = 0{,}2$ ✓; $C_T = 5\ \mu F$ ✓ (2)

[16]

QUESTION 5

5.1 Any TWO differences ✓✓✓✓: breadboard – no soldering, components plug in and can be reused, for testing / prototypes; Veroboard – components are soldered, copper strips must be cut to separate tracks, for permanent circuits; breadboard connections can work loose, Veroboard is more reliable. (4)

5.2 Any THREE in order ✓✓✓: plan the layout on paper / grid; place and solder the components; cut the copper tracks where needed; add wire links; check for solder bridges and test the circuit. (3)

5.3 Select the current (mA) range and plug the leads into the correct sockets ✓; break the circuit and connect the meter in series with the LED ✓. Precautions (any TWO) ✓✓: start on the highest range; never connect the ammeter across (in parallel with) a supply; observe polarity; switch off before connecting. (4)

5.4 $V_{pp} = 4 \times 2 = 8\ V$ ✓; $V_p = 8 \div 2 = 4\ V$ ✓; $T = 5 \times 1\ ms = 5\ ms$ ✓; $f = 1 \div T = 1 \div 0{,}005 = 200\ Hz$ ✓ (4)

[15]

QUESTION 6

6.1 NAND: 00→1, 01→1, 10→1, 11→0 ✓✓. XOR: 00→0, 01→1, 10→1, 11→0 ✓✓ (4)

6.2 $Y = A(\bar{B} + B) + \bar{A}B$ ✓ $= A + \bar{A}B$ ✓ $= A + B$ ✓ (3)

6.3.1 $Y = G \cdot (S + M)$ ✓✓ (2)

6.3.2 One 2-input OR gate ✓ and one 2-input AND gate ✓ (2)

6.4 Any TWO ✓✓: easy to change the program instead of rewiring; fewer wires and smaller; easier fault finding (status LEDs / monitoring); can do timing, counting and maths; more reliable (no moving contacts); can communicate with computers / networks. (2)

6.5 Receives the signals from switches and sensors and converts them into signals the processor can use (and isolates the processor) ✓ (1)

[14]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE